# S01 Foundations

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## Task A: Malware types and examples

| Type | Definition | Example + year | How it spread | Primary impact | Source |
|---|---|---|---|---|---|
| **Virus** | A computer virus is an ill-natured software application or authored code that can attach itself to other programs, self-replicate, and spread itself onto other devices. | PingPong, 1988 | The Ping-Pong virus spread through infected floppy disks. When users booted computers using infected disks, it infected the boot sector and subsequently spread to other inserted disks. | Its primary impact was displaying a bouncing ball on infected computer screens. Although generally harmless, it occasionally caused system crashes, particularly on certain older processors. | Grokipedia<br>En-Academic |
| **Worm** | A computer worm is standalone malicious software (malware) that automatically replicates itself and spreads between computers through networks, exploiting security vulnerabilities without requiring a host program to operate. | Morris Worm, 1988 | Computer worms spread through networks by exploiting security vulnerabilities, shared folders, email attachments, and infected websites. They automatically replicate themselves, infecting connected computers without requiring user interaction. | Computer worms consume network bandwidth, slow down systems, disrupt services, and may steal sensitive information, delete files, install backdoors, or cause widespread financial and operational damage. | Wikipedia: Computer worm |
| **Trojan** | A Trojan Horse is malicious software disguised as legitimate applications or files, tricking users into downloading it to gain unauthorized access, steal information, spy, or damage systems. | Zeus Trojan, 2007 | Trojans spread through deceptive emails, malicious attachments, fake software downloads, pop-up advertisements, and suspicious website links, tricking users into installing infected files on their computers. | Trojans can steal sensitive information, capture banking credentials, spy on users, provide unauthorized remote access, install additional malware, compromise privacy, and cause serious financial losses. | Fortinet |
| **Ransomware** | Ransomware is malicious software that encrypts files or locks computer systems, preventing users from accessing their data until attackers demand payment, usually in exchange for restoring access. | WannaCry Ransomware, 2017 | Ransomware spreads through phishing emails, malicious attachments, software vulnerabilities, stolen login credentials, infected websites, and compromised remote access systems, allowing attackers to infect computers and networks. | Ransomware encrypts important files, blocks system access, steals sensitive information, disrupts business operations, causes financial losses, damages organizational reputation, and forces victims to face ransom demands. | IBM |

| Type | Definition | Example + year | How it spread | Primary impact | Source |
|---|---|---|---|---|---|
| **Spyware** | Spyware is malicious software secretly installed on devices to monitor user activities, collect personal information, track browsing habits, steal passwords, and transmit sensitive data without users' knowledge. | Pegasus Spyware, 2016 | Spyware spreads through malicious email attachments, phishing links, infected websites, fake applications, pop-up advertisements, unsafe downloads, and outdated software containing security vulnerabilities that attackers exploit. | Spyware compromises user privacy, steals passwords and banking details, monitors online activities, collects personal information, enables identity theft, causes financial losses, and exposes confidential organizational data. | TechTarget |
| **Adware** | Adware is software that automatically displays unwanted advertisements on computers or mobile devices, often tracking users' browsing activities to generate advertising revenue or redirect them toward promotional websites. | Fireball, 2017 | Adware spreads through bundled software downloads, freeware installations, deceptive advertisements, malicious websites, browser extensions, and fake applications, often entering devices when users install seemingly legitimate programs. | Adware causes unwanted advertisements, browser redirects, slower device performance, privacy violations, excessive data consumption, tracking of browsing habits, and exposure to potentially harmful websites. | Check Point Research<br>Kaspersky |
| **Rootkit** | A rootkit is malicious software designed to hide its presence within a computer system, allowing attackers to maintain unauthorized access, control operations, and conceal other malicious activities. | Sony BMG copy protection rootkit, 2005 | Rootkits spread through infected software, malicious downloads, compromised applications, security vulnerabilities, and unauthorized installations, allowing attackers to secretly gain system privileges and maintain hidden access. | Rootkits conceal malicious activities, provide persistent unauthorized access, disable security protections, hide files and processes, compromise system integrity, and make malware detection and removal extremely difficult. | NIST<br>Virus Bulletin |
| **Keylogger** | A keylogger is software or hardware that secretly records keystrokes entered on a computer or mobile device, capturing passwords, messages, financial information, and other sensitive user data. | Agent Tesla, 2014 | Keyloggers spread through phishing emails, malicious attachments, infected software downloads, compromised websites, and unauthorized applications, secretly installing monitoring tools that record users' keyboard activities. | Keyloggers steal passwords, banking credentials, private messages, and confidential information, enabling identity theft, financial fraud, account hijacking, privacy violations, and unauthorized access to personal accounts. | MITRE ATT&CK<br>BTH (DiVA) |
| **Botnet agent** | A botnet agent is malicious software that infects devices and connects them to a network controlled by cybercriminals, allowing attackers to remotely control devices and perform coordinated activities. | Mirai, 2016 | Botnet agents spread through weak passwords, unpatched security vulnerabilities, malicious downloads, phishing emails, and infected devices, connecting compromised computers and IoT devices to centralized attacker-controlled networks. | Botnet agents enable distributed denial-of-service attacks, spam distribution, data theft, unauthorized computing activities, network disruption, financial losses, and remote control of thousands of compromised devices. | Microsoft: Mirai |
| **Fileless / living-off-the-land** | Fileless malware operates mainly in computer memory, abusing legitimate system tools and processes instead of traditional malicious files, making detection and removal more difficult for security software. | Astaroth, 2018 | Fileless malware spreads through phishing emails, malicious documents, compromised websites, and exploited vulnerabilities, abusing legitimate system utilities to execute malicious commands without traditional executable files. | Fileless malware steals credentials, collects sensitive information, bypasses traditional antivirus detection, maintains unauthorized access, compromises systems, disrupts operations, and makes forensic investigation significantly more challenging. | Microsoft: Astaroth |

| Type | Definition | Example + year | How it spread | Primary impact | Source |
|---|---|---|---|---|---|
| **Logic bomb** | A logic bomb is malicious code secretly inserted into software that activates when specific conditions or events occur, potentially deleting files, disrupting systems, or damaging organizational data. | Omega Engineering Logic Bomb, 1996 | Logic bombs are introduced through malicious software modifications, insider access, compromised applications, or unauthorized code changes, remaining hidden until predefined conditions trigger their execution within systems. | Logic bombs can delete critical files, corrupt databases, interrupt business operations, damage software systems, cause financial losses, destroy important information, and disrupt organizational productivity significantly. | PBS Frontline |
| **Bootkit** | A bootkit is malicious software that infects a computer's boot process or firmware, allowing it to load before the operating system and maintain stealthy, persistent control over the infected device. | TDL4, 2011 | Bootkits spread through infected software downloads, malicious email attachments, exploited security vulnerabilities, compromised applications, and infected external devices, gaining access to systems and modifying their boot processes. | Bootkits compromise system startup, hide malicious activities, bypass security protections, maintain persistent unauthorized access, disable security software, steal sensitive information, and make malware removal extremely difficult. | WeLiveSecurity (ESET) |

### Most dangerous malware for a Pakistani SME

Ransomware is arguably the most dangerous malware for a Pakistani SME because it can encrypt important business data, interrupt daily operations, and cause significant financial losses. Pakistani SMEs may have limited cybersecurity budgets and weaker security systems, making recovery from such attacks particularly difficult. Therefore, businesses should use updated antivirus software, strong passwords, regular data backups, employee cybersecurity training, and multi-factor authentication to reduce these risks

# Task B

## B2: Linux file permissions

**Goal:** create a file, change permissions three times (mix symbolic and octal), explain each.

### Step 1: Create the file

```
mkdir -p ~/b2_lab
cd ~/b2_lab
echo 'permissions practice' > test.txt
ls -l test.txt
stat -c '%a %A' test.txt
```

For an ordinary file, the three octal digits represent owner, group, and other permissions. The values in each digit are added: rwx = 4 + 2 + 1 = 7, rw- = 4 + 2 = 6, r-- = 4, and --- = 0.

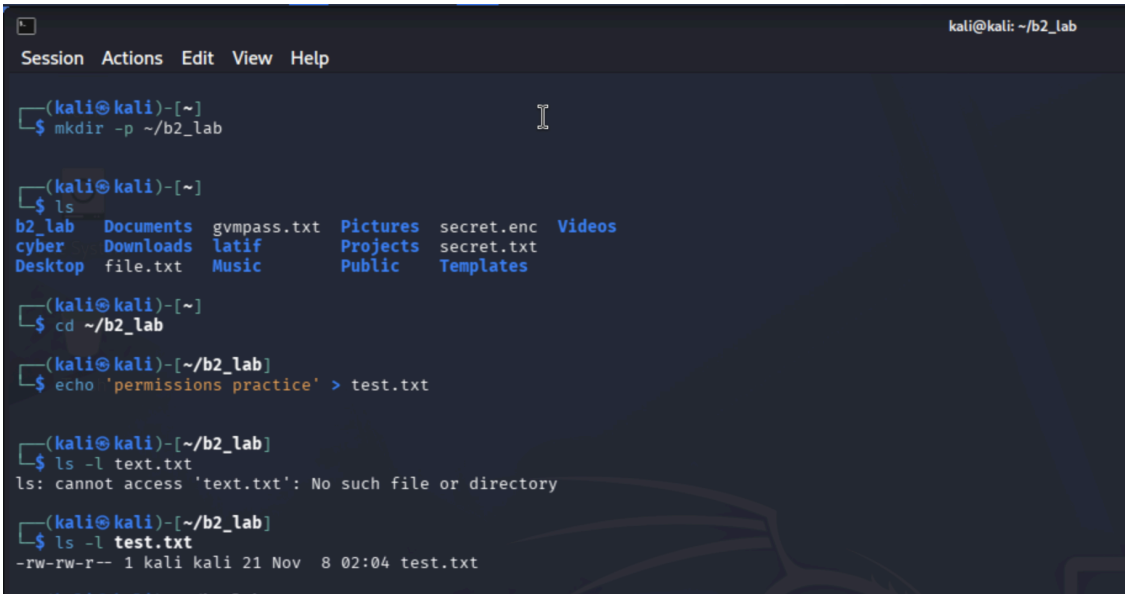



**Figure 1:** *Creating the b2_lab directory and test.txt, then listing it with ls -l (the first attempt, ls -l text.txt, was a typo and failed).*




**Figure 2:** *stat output showing the starting permissions of test.txt: 664 (-rw-rw-r--).*

### Step 2: Change 1 (octal)

```
chmod 600 test.txt
ls -l test.txt
```

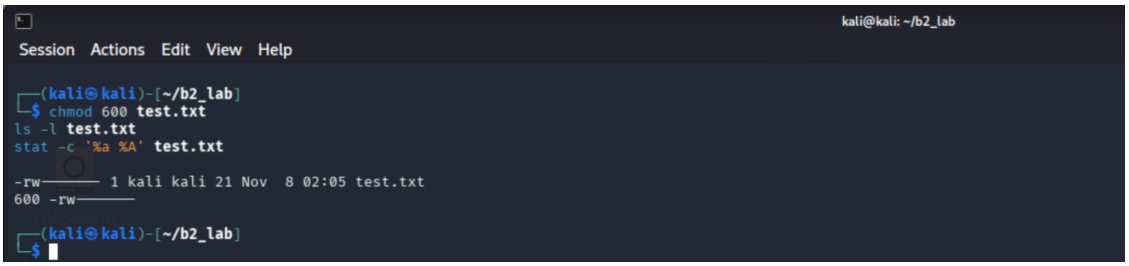



**Figure 3:** *chmod 600 test.txt; the permissions change to -rw------- (600).*

Explanation: 600 gives the owner read and write permission (6 = 4 + 2), while the group and other users receive no permissions (0). Setting an octal mode replaces all of the existing permission bits at once, rather than adding or removing a single permission.

## Step 3: Change 2 (symbolic)

```
chmod g+r test.txt
ls -l test.txt
```

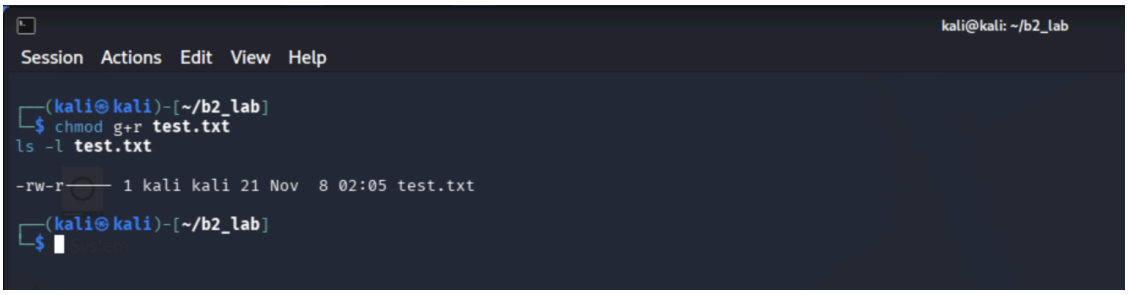



**Figure 4:** *chmod g+r test.txt; the permissions change to -rw-r----- (640).*

Explanation: g+r adds read permission for the group while retaining the existing owner and other permissions.

## Step 4: Change 3 (octal)

```
chmod 755 test.txt
ls -l test.txt
```




**Figure 5:** *chmod 755 test.txt; the permissions change to -rwxr-xr-x (755).*

Explanation: 7 for the owner means read + write + execute (4 + 2 + 1); 5 for the group means read + execute (4 + 1); and 5 for others means read + execute (4 + 1).

### Summary Table B2

| Step | Command | Octal | Symbolic result | What changed |
|---|---|---|---|---|
| Start | echo > test.txt | 664 | -rw-rw-r-- | Owner = read, write<br>Group = read, write<br>Others = read |
| 1 | chmod 600 | 600 | -rw------- | Owner = read, write<br>Group = none<br>Others = none |
| 2 | chmod g+r | 640 | -rw-r----- | Owner = read, write<br>Group = read<br>Others = none |
| 3 | chmod 755 | 755 | -rwxr-xr-x | Owner = read, write, execute<br>Group = read, execute<br>Others = read, execute |

## B4: Encrypt with OpenSSL and decode the header with xxd

**Goal:** encrypt a file, look at the first 16 bytes, explain Salted__.

### Step 1: Create a secret file

```
mkdir -p ~/cyber && cd ~/cyber
echo 'my secret message' > secret.txt
cat secret.txt
```

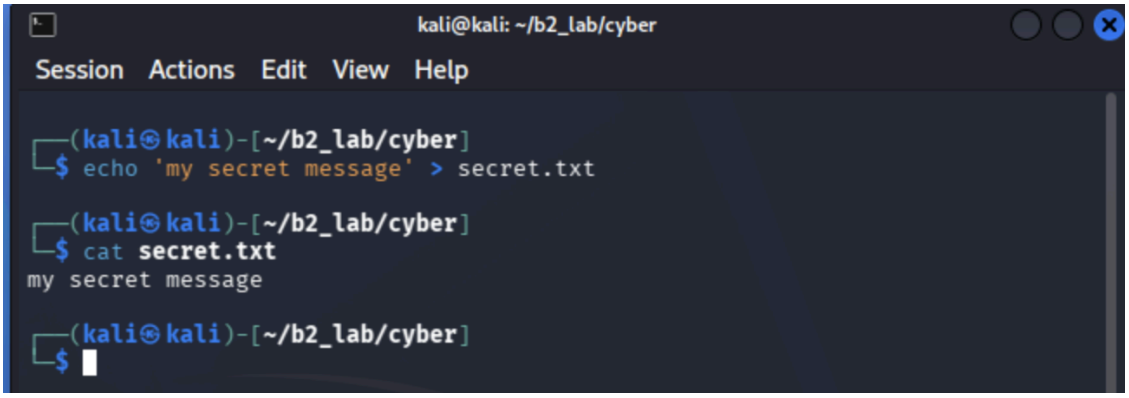



**Figure 6:** *Creating secret.txt and displaying its contents with cat.*

### Step 2: Encrypt

```
openssl enc -aes-256-cbc -salt -pbkdf2 -in secret.txt -out secret.enc
ls -l secret.txt secret.enc
```




**Figure 7:** *Encrypting secret.txt with AES-256-CBC; secret.enc is 48 bytes and secret.txt is 18 bytes.*

### Step 3: Look inside

```
xxd -l 16 secret.enc
file secret.enc
```

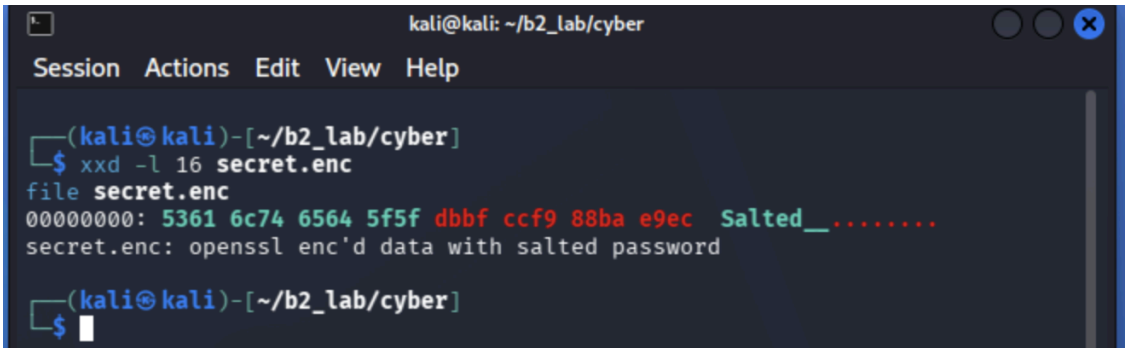



**Figure 8:** *The first 16 bytes of secret.enc in hex; the ASCII column shows Salted__ and the file command reports salted OpenSSL data.*

Explanation: the first 8 bytes (53 61 6c 74 65 64 5f 5f) are the ASCII text "Salted__", a magic header that OpenSSL writes to show the file was encrypted with a salted password. The next 8 bytes (db bf cc f9 88 ba e9 ec) are the random salt, which is combined with the password to derive the key, so the same password gives a different key and different ciphertext each time. The remaining 32 bytes are the encrypted data (the 18-byte message padded to two 16-byte AES blocks), giving the 48-byte file.

### Step 4: Decrypt to prove it works

```
openssl enc -d -aes-256-cbc -pbkdf2 -in secret.enc -out decrypt.txt
cat decrypt.txt
```

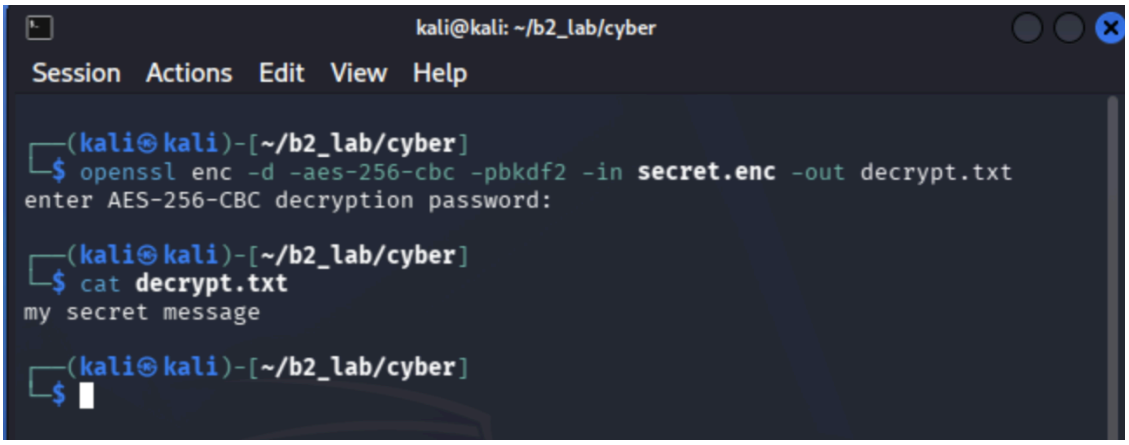



**Figure 9:** *Decrypting secret.enc with the same password; decrypt.txt shows the original message.*

## GitHub Repository

```
digital-forensic-cyber-security-pseb-bootcamp/
└── s01-foundations/
    ├── README.md
    └── Figures/
        └── screenshots.jpg
```